Language-model based investigation of information structure in code-switched dialogue

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Background

Code-switching (CS)

Code-switching, i.e., **alternation of multiple languages** in the same discourse or conversation, is primarily a dialogue phenomenon.

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Language choice in code-switched interaction has been analysed as a function of the underlying linguistic structures (Kootstra, Dijkstra, and Van Hell, 2020), speaker proficiency, and language identity of the speaker (Jan-Petter and Gumperz, 2020).

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Recent studies argue that **code-switching increases communicative efficiency** in multilingual interaction and it is used to emphasise important information (Beatty-Martínez, Navarro-Torres, and Dussias, 2020).

Code-switching (CS)

Consider the following example:

(1) A. *yeah but tú sabes qué?*

yeah but you know what?

Code-switching (CS)

Consider the following example:

(2) A. *yeah but tú sabes qué?*

yeah but you know what?

A. *el problema de aquí que no me gustan son los taxes is too high (...)*

the problem that I dislike over here is that taxes are too high

Code-switching (CS)

Consider the following example:

(3) A. *yeah but tú sabes qué?*

yeah but you know what?

A. *el problema de aquí que no me gustan son los taxes is too high (...)*

the problem that I dislike over here is that taxes are too high

B. *I see (...)*

Code-switching (CS)

Consider the following example:

(4) A. *yeah but tú sabes qué?*

yeah but you know what?

A. *el problema de aquí que no me gustan son los taxes is too high (...)*

the problem that I dislike over here is that taxes are too high

B. *I see (...)*

B. *siempre por lo general el impuesto es del dos por ciento*

generally, tax is always at 2 per cent

Code-switching (CS)

Consider the following example:

(5) A. *yeah but tú sabes qué?*

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A. *el problema de aquí que no me gustan son los taxes is too high (...)*

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generally, tax is always at 2 per cent
(Zeledon5, Miami-Bangor; Deuchar, 2010)

Code-switching in dialogue

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Meaning predictability in CS

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- The predictability of the code-switch depends on both **speaker-internal** and **speaker-external** factors
- A study by Myslín and Levy (2015) investigating both factors concluded that **contextual (i.e., speaker-external) factors such as participant constellation and meaning predictability predict code-switching** better than speaker-internal factors.

Our study

The broad aim of this study is:

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- To investigate how information is distributed in relation to code-switching in interaction

Research Hypotheses

In this study, we tested the following information-theoretic hypotheses on code-switching in discourse (Myslín and Levy, 2015):

- Multilingual speakers switch languages at lower information points in dialogue

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- Multilingual speakers switch languages at lower information points in dialogue
- High information content is always signalled using the same language in bilingual dialogue
- Position of the switch within the utterance (and discourse) has an effect on information value

Methodology

Corpus

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Using the corpus, we measured:

1. Surprisal at code-switching points within utterances
2. Informativeness of the immediate dialogue context

Sampling from the corpus

We created balanced sets of mid-utterance code-switch points (n=800) and analogous points with no code-switch (n=800). These sets were created by sampling tokens as follows:

- used transcripts that had at least 25 eligible CS points
- excluded the first token and the final 5 tokens in each utterance
- excluded the first 50 utterances
- excluded separators, pauses, word fragments and other non-words
- in the CS case: the previous token is from another language

Language model-based surprisal metrics

- We use **information value (IV)** (Giulianelli, Wallbridge, and Fernández, 2023a) as a proxy for (human) surprisal
    - **IV intuition:** Given some context, LM generations will be more similar to the actual continuation when the continuation is less surprising (given the context).
    - IV has been shown to correlate better with human surprisal when compared to more traditional LM-based metrics like perplexity (Giulianelli, Wallbridge, and Fernández, 2023a)

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  - IV has been shown to correlate better with human surprisal when compared to more traditional LM-based metrics like perplexity (Giulianelli, Wallbridge, and Fernández, 2023a)
- We measure the IV (surprisal) of the continuation *with* dialogue context taken into account

Example LM-based surprisal metrics

@Begin
@Languages: eng, spa
@Participants: CHA Chantal Teenager, GIL Gillian Child
@Situation: Informal conversation between two cousins
@Date: 22-MAR-2008
GIL: ah Jacky está en Colombia para Navidad un año.
GIL: um yo (.) fui la día primero un poquito días de Navidad.
GIL: i and then um I stayed there for a week or like couple days
or something like...
GIL: I came like a week before Christmas you know.
. . .
CHA: what are you favourite hobbies?
GIL: I u draw.
GIL: I sing.
GIL: and sometimes I go play golf.
CHA: muy chévere.
CHA: my hobbies are— I—
GIL: and swimming.
CHA: I— I love to play basketball.
CHA: I'm in my team in my school.
CHA: um I also love to swim.
CHA: you know in the pool it's very very fun.
GIL: I know.
GIL: remember the other day when we— (.) cuando nosotros fuimos a la piscina aquí?

when we went to the pool herae?

We always include the first part of the transcript for minimal context and to expose the model to the transcription format.

This is the sampled point (in this case a CS point)

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GIL: I know.
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Actual continuation:
cuando nosotros fuimos a la piscina aquí?
when we went to the pool here?

Generated with dialogue context:

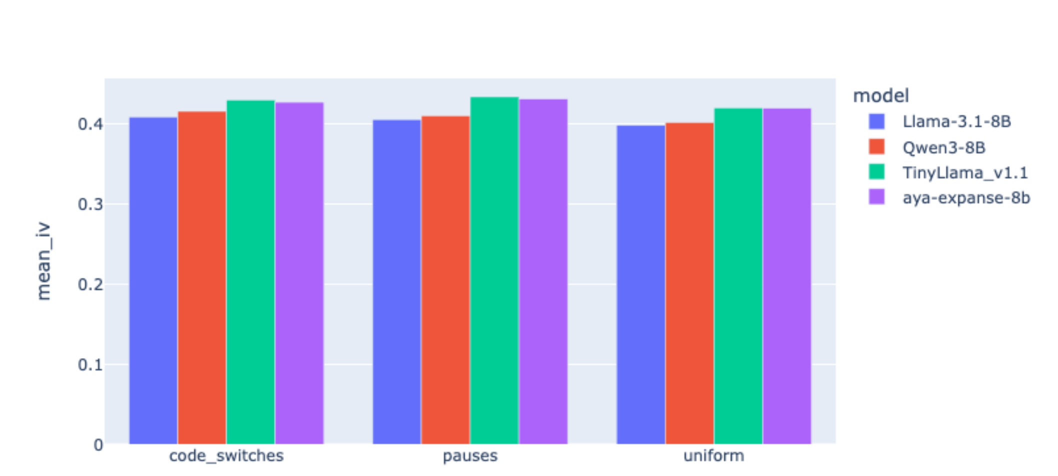
1. *we went to the pool.* **0.20**
2. *when we were playing basketball?* **0.35**
3. *when we went swimming?* **0.30**

IV_diag
0.30

mean cosine distance from actual continuation

IV across Language Models

To find the best Language Model to use for our experiment, we sampled IV using four different LMs and measured IVs in three different contexts: Code-switches, Pauses, and Uniform (no CS and pauses)



Analysis and Results

Statistical Analysis

To test whether code-switching has an effect on information value (IV) derived from Llama 3.1, we fitted a mixed effects model (lmer; Bates et al., 2015) with Code-switching (True/False), Continuation Language, Context Size, Token position in turn and Utterance position as the predictors and IV as the outcome.

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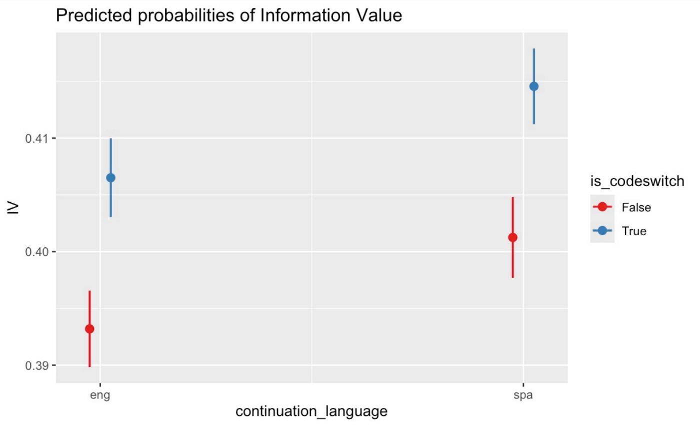
The results show that **information value increases** when there is a code-switch

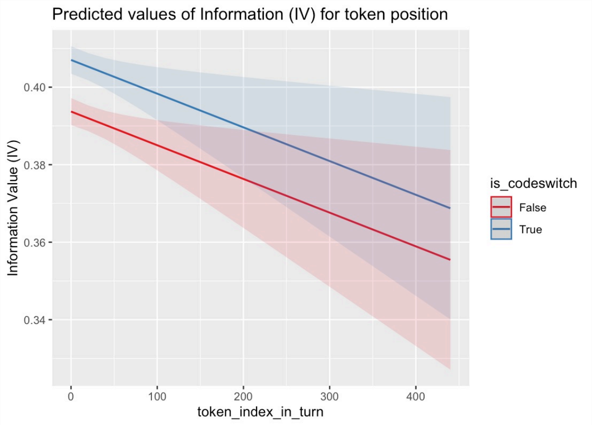
Interestingly, **ENG-SPA switches resulted in higher information value** compared SPA-Eng switches.

Results: Higher Information Value at CS points



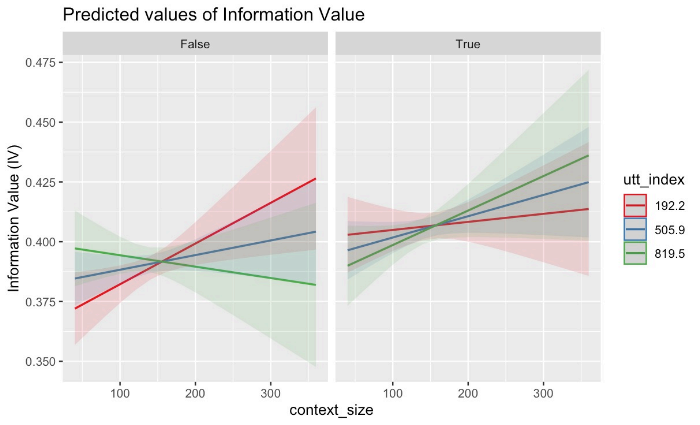
Information value increases at code-switching points, especially when the language switches from English to Spanish

Results: Does surprisal vary by position of the switch?



The token position of the switch has a small effect on information value.)

Results: Does surprisal vary by position of the switch?



The utterance position of the switch did not have a significant effect on IV. However, IV seems to increase when there is a switch towards the end of the dialogue (i.e., higher utt-index)

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Conclusion

- LLMs demonstrate surprisal similar to human surprisal in code-switched interaction.
- In spontaneous interaction, code-switching is more likely to occur at high-information points (i.e., high surprisal) and it is often signalled using Spanish in Spanish-English bilinguals in Miami, Florida.
- **Code-switching is used to incrementally update information in interaction**.

Future Directions

- Test whether information distribution varies between language pairs/communities

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- Test whether information distribution varies between language pairs/communities
- Analyse the effect of number of prior turns on surprisal in code-switched dialogue.
- Measure information value in relation to syntactic dependencies in dialogue

Questions?

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